Night shift — extending matched SHD training

ar070 · 19 July 2026 · Draft

Digest

Empty until the registered experiment finishes.

Mandate

Purpose

`exp068` produced a promising one-seed PING advantage, while PING's best validation checkpoint occurred at the final allowed epoch. This night tests the narrowest explanation: the matched recipe may simply be under-trained at forty epochs. It changes training duration only. It is exploratory and does not estimate a population-level architecture effect.

Registered experiment

Field	exp069
Hypothesis	Under the unchanged exp068 recipe, extending matched training from 40 to 80 epochs raises PING's selected held-out validation accuracy by at least 3 percentage points above its 43.50% exp068 baseline.
Baseline	exp068 selected COBA epoch 25 at 40.81% and PING epoch 40 at 43.50% on the identical 816-utterance validation split. Its official-test results are context only and are not outcomes for exp069.
Kill criterion	Kill either cell for non-finite loss or logits, persistent skipped updates, silence, clear saturation, or failure to complete the fixed protocol; kill the comparison for different partitions, mismatched registered settings, architecture-specific tuning, or any access to the official SHD test split.
Seed count	1
Budget	One local plumbing smoke plus one 80-epoch full run per architecture; two concurrent pods maximum, 4 h backstop per pod, and 10 USD hard total RunPod spend.
Status	queued

Fixed data and settings

Reuse exp068′s deterministic seed-42 joint speaker/class split: 7,340 development-training and 816 validation utterances, with identical index hashes in COBA and PING. The official SHD test split must not be staged, loaded, evaluated, or inspected. Repeated official-test use during exploratory optimization would turn it into a development target.

Preserve every exp068 setting except the registered duration change: 256 excitatory and 64 inhibitory cells, fixed Dale-constrained recurrence, input-weight mean 0.9, 95% input sparsity, membrane-mean readout scaled by 225, Adam learning rate 0.0004, batch size 32, 1 ms timestep, 1,000 ms duration, and no firing-rate regularizer. COBA disables the inhibitory loop and uses no voltage-gradient dampening; PING enables the loop and uses dampening 1000. Initialisation, ordering, inputs, readout, and all other settings remain matched. No architecture-specific tuning or rescue is permitted.

Training, outcomes, and interpretation

Run a local 128-training/128-validation, two-epoch plumbing smoke first. If both cells remain finite and active, train each full cell for exactly 80 epochs. Select the checkpoint with highest validation accuracy; break ties by lower validation cross-entropy and then earlier epoch.

The primary outcome is PING's selected validation-accuracy change relative to exp068′s 43.50% baseline. At least +3 percentage points is a useful longer- training signal; a positive gain below +3 points is weak; zero or negative gain provides no evidence that forty epochs caused the prior ceiling.

Registered diagnostics are COBA's change from 40.81%, the contemporaneous PING-minus-COBA validation difference, full learning and activity curves, selected epoch and loss, E/I firing rates, skipped and non-finite batches, runtime, peak GPU memory, and matched validation input/E/I rasters. None is a substitute primary outcome.

Operating contract


```
budgets:
  runpod_total_usd: 10
  max_pods_concurrent: 2
  wall_clock_backstop_per_pod: 4h
  seeds_default: 1
scope:
  collection: spiking-heidelberg-digits
  experiment_id: exp069
  activity_trace: ar071
  may_edit_snn_cli: false
  official_test_access: forbidden
stop_when:
  - queue_empty
  - runpod_budget_exhausted
  - protocol_integrity_failure
  - build_red_twice
```


Use `experiments/exp069.py` and existing CLI capabilities. Do not edit `tools/snn`. Use focused lightweight checks and `demolab build`; do not run the full test suite on the local 4 GB host. Two concurrent pods, one per architecture, are authorized within the 10 USD hard ceiling. Reap them after completion or failure. Stop for authority before changing this design, exceeding the budget, or taking an action outside this scope.

Record

2026-07-19 00:45–00:49 UTC: publication accepted and mandate approved

The scientist accepted exp068 by merging pull request 51. Its merge commit is `569926f`. The next experiment was then narrowed to training duration alone, with validation-only outcomes so the already observed official test does not become an optimization target. The scientist approved this complete contract at 00:48:51 UTC. No exp069 runner, compute, result, or cost existed at approval.

2026-07-19 00:51 UTC: mandate anchor and night branch

The approved mandate and both reserved article identifiers were committed to `main` at `3c440aa`. The dedicated night branch was created from that exact anchor. At branch creation, exp069 spend remained 0 USD, no pod was active, and the official SHD test remained outside the registered experiment scope.

2026-07-19 01:35–01:40 UTC: implementation and smoke gate

The validation-only runner was implemented without changing `tools/snn`. Unlike exp068, it contains no official-test download, staging, loading, or evaluation route; its matched rasters use fixed validation examples. Focused Ruff, type, and compilation checks passed. The registered 128-training / 128-validation, two-epoch smoke passed for both cells with finite losses, zero skipped or non-finite updates, active COBA E firing at 23.93 Hz, and active PING E/I firing at 2.95 / 14.56 Hz. The derived development split hashes exactly matched exp068. No pod had been created and spend remained 0 USD at the smoke gate.

2026-07-19 01:42–01:43 UTC: full-cell dispatch

Commit `b944ebf` pinned the passing smoke implementation before compute. The dispatch dry-run resolved to exactly one COBA and one PING cell. Two single-GPU pods were then created concurrently at 0.99 USD/hour each with four-hour backstops. One transient capacity miss on the PING request was resolved by the registered provisioning retry; no extra pod was created. The verified fleet count was two, and worst-case exposure was 7.92 USD.